

Full solution packet for arXiv:2501.17505

A Hardy-block formula for the missing associate norm

Status. full_solution_likely_valid. This packet answers the printed associate-space problem in item (iii), p. 36, of Miquel Saucedo and Sergey Tikhonov, *The Fourier transform is an extremizer of a class of bounded operators*, arXiv:2501.17505. In the Banach range $1 \leq q < 2$, it identifies the requested associate functional as an explicit block space and gives its norm as an attained minimum.

Source problem (item (iii), p. 36). After finding the optimal Fourier-domain functional, the source says that it is of interest to find its associate. It records a known formula for the simpler of two functionals and states: “We have not been able to find an expression for the latter in the literature.”

and the left-hand side norm is optimal.

(iii) *As the previous example shows, in light of Corollary 5.3, it is of interest to find the associates of the norms*

$$\left(\int_0^\infty u^{*,q}(t) \left(\int_0^{t^{-1}} F^* \right)^q dt \right)^{\frac{1}{q}}$$

and

$$\left(\int_0^\infty U^{\frac{q^\sharp}{2}}(t) u^{*,q}(t) \xi^{-\frac{q^\sharp}{2}}(t) t^{-\frac{q}{2}} \left(\int_0^t \left(\int_0^{r^{-1}} F^* \right)^2 dr \right)^{\frac{q}{2}} dt \right)^{\frac{1}{q}},$$

as these yield the sharpest rearrangement-invariant inequalities of the type
 $\left\| \widehat{f} \right\|_X \lesssim \|f u^{-1}\|_{q'}$ *for a given* u .

*By [17, Remark 6.3], the associate of the former is given by $\|w f^{**}\|_{q'} < \infty$ with $w(t) = U^{-1/q}(1/t) t^{-1/q'}$. We have not been able to find an expression for the latter in the literature.*

The functional from the source

Fix $1 \leq q < 2$ and set

$$\begin{aligned} \frac{1}{q^\sharp} &= \frac{1}{q} - \frac{1}{2}, & U(t) &= \int_0^t u^*(s)^q ds, \\ \xi(t) &= U(t) + t^{q/2} \left(\int_t^\infty u^*(s)^{q^\sharp} ds \right)^{q/q^\sharp}. \end{aligned}$$

Write

$$a(t) = U(t)^{q^\sharp/2} u^*(t)^q \xi(t)^{-q^\sharp/2} t^{-q/2}. \tag{1}$$

The second functional displayed on p. 36 is

$$\mathcal{N}_a(F) = \left(\int_0^\infty a(t) \left[\int_0^t \left(\int_0^{1/r} F^*(s) ds \right)^2 dr \right]^{q/2} dt \right)^{1/q}. \tag{2}$$

As in Definition 1.1 of the source, its associate is

$$\mathcal{N}'_a(G) = \sup \left\{ \int_0^\infty |F(s)G(s)| ds : \mathcal{N}_a(F) \leq 1 \right\}. \quad (3)$$

Result at a glance

Let

$$\Omega = \{(t, r) : 0 < r < t < \infty\}$$

and, for a measurable kernel K on Ω , define

$$\|K\|_{q',a} = \left(\int_0^\infty a(t) \left[\int_0^t |K(t,r)|^2 dr \right]^{q'/2} dt \right)^{1/q'} \quad (4)$$

when $q > 1$. At $q = 1$, (4) means

$$\|K\|_{\infty,a} = \operatorname{esssup}_{t:a(t)>0} \left(\int_0^t |K(t,r)|^2 dr \right)^{1/2}. \quad (5)$$

Define the positive adjoint Hardy operator

$$(\mathcal{T}_a^* K)(s) = \int_0^\infty a(t) \int_0^{\min\{t, 1/s\}} K(t,r) dr dt. \quad (6)$$

For nonnegative functions g, h on $(0, \infty)$, write

$$g \preceq_H h \iff \int_0^x g(s) ds \leq \int_0^x h(s) ds \quad \text{for every } x > 0. \quad (7)$$

Then the requested associate has the exact formula

$$\boxed{\mathcal{N}'_a(G) = \min \{ \|K\|_{q',a} : K \geq 0, \quad G^* \preceq_H \mathcal{T}_a^* K \}.} \quad (8)$$

Thus the associate is a concrete Hardy-block space. Expanding the constraint in (8), admissibility of K means

$$\int_0^x G^*(s) ds \leq \int_0^\infty a(t) \int_0^t K(t,r) \min\{x, 1/r\} dr dt \quad (x > 0). \quad (9)$$

Formula (8) is valid with value $+\infty$ as well. Whenever the associate norm is finite, a minimizing kernel exists.

Proof intuition

The nested integral in (2) looks nonlinear only because an L^2 norm is written out. On decreasing functions it is the norm of the positive linear map

$$(\mathcal{T}f)(t, r) = \mathbf{1}_{\{r < t\}} \int_0^{1/r} f(s) ds$$

into the mixed space $L^q(a(t)dt; L^2(0, t))$. Dualizing that mixed space introduces the kernel K in (4). Fubini's theorem gives the explicit adjoint (6).

The only remaining feature is that an associate of a rearrangement-invariant functional tests decreasing rearrangements. A function h dominates g against every nonnegative decreasing test function exactly when $g \preceq_H h$. Consequently, the adjoint need not equal G^* pointwise; it must dominate it in the cumulative Hardy order. A separation argument proves that this necessary condition is also sufficient and that no norm is lost.

Formal theorem and proof

The proof works for a general locally integrable weight $a \geq 0$. We assume only the natural nondegeneracy condition that (2) is finite on bounded, compactly supported decreasing functions; this holds in every nontrivial case in which the source functional defines an associate space.

Lemma 1 (The Hardy cone). *Let $p \in L^1_{\text{loc}}(0, \infty)$. Then*

$$\int_0^\infty f(s)p(s) ds \geq 0$$

for every nonnegative, nonincreasing, compactly supported step function f if and only if

$$\int_0^x p(s) ds \geq 0 \quad (x > 0). \quad (10)$$

Consequently, for nonnegative g, h ,

$$\int_0^\infty fg \leq \int_0^\infty fh \quad \text{for all nonnegative decreasing } f$$

if and only if $g \preceq_H h$.

Proof. Necessity follows by taking $f = \mathbf{1}_{(0,x)}$. Conversely, every nonnegative decreasing step function has the form

$$f = \sum_{j=1}^m c_j \mathbf{1}_{(0,x_j)}, \quad c_j \geq 0.$$

Equation (10) then gives $\int fp \geq 0$. General decreasing tests follow by monotone approximation. Apply this with $p = h - g$ for the final assertion. $\square$

Theorem 2 (Associate of the nested Hardy functional). *Let $1 \leq q < \infty$ and let $a \geq 0$ be locally integrable. Define*

$$\mathcal{N}_a(F) = \left(\int_0^\infty a(t) \left[\int_0^t \left(\int_0^{1/r} F^*(s) ds \right)^2 dr \right]^{q/2} dt \right)^{1/q}.$$

If $\mathcal{N}_a$ is finite on compactly supported decreasing step functions, then its associate is given by (8)–(9), with (5) when $q = 1$. If $\mathcal{N}'_a(G) < \infty$, the minimum in (8) is attained.

Proof. Put

$$Z_q = L^q((0, \infty), a(t)dt; L^2(0, t)).$$

For $q = 1$ its Banach dual is $Z_\infty = L^\infty(a(t)dt; L^2(0, t))$; for $q > 1$ its dual is $Z_{q'}$. Their duality pairing is

$$\langle H, K \rangle = \int_0^\infty a(t) \int_0^t H(t, r) K(t, r) dr dt. \quad (11)$$

By the Hardy–Littlewood rearrangement inequality, (3) becomes

$$\mathcal{N}'_a(G) = \sup \left\{ \int_0^\infty f(s)G^*(s) ds : f \geq 0 \text{ decreasing, } \|\mathcal{T}f\|_{Z_q} \leq 1 \right\}, \quad (12)$$

where

$$(\mathcal{T}f)(t, r) = \mathbf{1}_{\{0 < r < t\}} \int_0^{1/r} f(s) ds. \quad (13)$$

Indeed, $\|\mathcal{T}f\|_{Z_q} = \mathcal{N}_a(f)$ by inspection.

For $K \in Z_{q'}$, Tonelli's theorem gives

$$\begin{aligned} \langle \mathcal{T}f, K \rangle &= \int_0^\infty a(t) \int_0^t K(t, r) \int_0^{1/r} f(s) ds dr dt \\ &= \int_0^\infty f(s) (\mathcal{T}_a^* K)(s) ds, \end{aligned} \quad (14)$$

first for nonnegative functions and then by decomposition. This proves the adjoint formula (6).

Suppose $K \geq 0$ and $G^* \preceq_H \mathcal{T}_a^* K$. Lemma 1, (14), and mixed-norm Hölder give, for every decreasing $f \geq 0$,

$$\int f G^* \leq \int f \mathcal{T}_a^* K = \langle \mathcal{T}f, K \rangle \leq \|\mathcal{T}f\|_{Z_q} \|K\|_{q', a}.$$

Taking the supremum in (12) proves

$$\mathcal{N}'_a(G) \leq \inf_{K \geq 0, G^* \preceq_H \mathcal{T}_a^* K} \|K\|_{q', a}. \quad (15)$$

We prove the reverse inequality by separation. Let $C = \mathcal{N}'_a(G) < \infty$, and let $\mathcal{P}$ be the cone of locally integrable p satisfying (10). In the weak local topology generated by compactly supported decreasing step functions, consider

$$\mathcal{B}_C = \{\mathcal{T}_a^* K - p : \|K\|_{q', a} \leq C, p \in \mathcal{P}\}. \quad (16)$$

The dual ball is weakly compact when $q > 1$ and weak-star compact when $q = 1$; $\mathcal{P}$ is weakly closed by (10). Since the sum of a compact set and a closed set is closed, $\mathcal{B}_C$ is closed. These statements may equivalently be verified first on finite intervals and finite step partitions, then passed to the limit by weak compactness.

If $G^* \notin \mathcal{B}_C$, Hahn–Banach separation gives a compactly supported test function f such that

$$\int f G^* > \sup_{\|K\|_{q', a} \leq C, p \in \mathcal{P}} \left(\int f \mathcal{T}_a^* K - \int f p \right). \quad (17)$$

The supremum on the right can be finite only if f belongs to the polar of $\mathcal{P}$. By Lemma 1, this means that f is nonnegative and decreasing. Equations (11)–(14) then show that the right side of (17) is

$$C \sup_{\|K\|_{q', a} \leq 1} \langle \mathcal{T}f, K \rangle = C \|\mathcal{T}f\|_{Z_q}.$$

This contradicts (12) and the definition of C . Hence $G^* = \mathcal{T}_a^* K - p$ for some $\|K\|_{q', a} \leq C$ and $p \in \mathcal{P}$, or equivalently

$$G^* \preceq_H \mathcal{T}_a^* K.$$

Replacing K by $|K|$ preserves its norm and only increases $\mathcal{T}_a^* K$, so we may take $K \geq 0$. Together with (15), this proves equality and also attainment of the minimum.

Finally, integrating (6) over $s \in (0, x)$ and applying Tonelli yields

$$\int_0^x (\mathcal{T}_a^* K)(s) ds = \int_0^\infty a(t) \int_0^t K(t, r) \min\{x, 1/r\} dr dt,$$

which is exactly (9). □

Corollary 3 (Answer to the problem on p. 36). *Let $1 \leq q < 2$, let u be as in Corollary 5.3 of Saucedo–Tikhonov, and assume $\xi(t) < \infty$ for $t > 0$. Form a from (1). The associate of the second functional in item (iii), p. 36, is the Hardy-block functional in (8), equivalently (4)–(5) subject to the explicit inequalities (9). Therefore the sharp rearrangement-invariant target norm requested there is completely determined by an attained convex minimization problem.*

Proof. Substitution of (1) into Theorem 2 gives exactly the second displayed functional on p. 36. Definition 1.1 of the source is (3), so the theorem computes the requested associate. $\square$

Checks, interpretation, and scope

Consistency check. When $\xi \asymp U$, Corollary 5.3 of the source simplifies (2), up to constants depending only on q , to the first Hardy functional displayed on p. 36. The source cites the known associate $\|wG^{**}\|_{q'}$ for that functional. Since equivalent functionals have equivalent associates, (8) agrees with that known formula in the simplifying regime. Formula (8) additionally covers the genuinely harder case in which ξ/U is not uniformly bounded.

What the formula provides. The original supremum in (3) ranges over all input functions. Formula (8) replaces it by a convex mixed-norm minimization over positive kernels, with the elementary linear constraints (9). On a finite grid it is a second-order cone program. More importantly, (8) identifies the functions in the associate space and their norm without reference to Fourier transforms or to an unspecified best constant.

Limitations. The answer is a two-variable block representation, not in general a single weighted Lorentz or Γ norm. No claim is made that the kernel can be eliminated for arbitrary irregular u . The theorem covers the Banach associate range $1 \leq q < 2$, including $q = 1$; it does not assert a Banach-dual formula for $0 < q < 1$.

Novelty check. The run indexes were searched for the arXiv id, title, associate norms, Hardy/Copson duality, and the defining nested integral. Current arXiv searches used the exact printed phrase, the authors, the displayed U – ξ structure, Köthe associates of mixed Hardy/Copson spaces, and papers citing arXiv:2501.17505. They found the source paper and general Cesàro/Copson duality literature, but no answer to this exact functional and no later paper citing the source with such an answer. Novelty confidence is moderate: the conic-polar mechanism is standard, but this exact associate identification and its application to the source functional were not found.

Human-review recommendation. Review as a full answer in block-space form. The main verifier focus should be the closedness/separation step leading from (12) to (16), especially the weak-star endpoint $q = 1$. A second useful check is whether a preferred definition of “expression” in this subliteration demands a one-variable weighted norm; mathematically, (8) computes the associate and its exact best constant, but it deliberately retains an auxiliary kernel.

References

- [1] M. Saucedo and S. Tikhonov, *The Fourier transform is an extremizer of a class of bounded operators*, arXiv:2501.17505, especially Definition 1.1, Corollary 5.3, and item (iii) on p. 36.